

Exploring Interaction Paradigms for Performing Medical Image Segmentation Tasks in Virtual Reality

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ABSTRACT

Virtual reality (VR) facilitates immersive visualization and interaction with complex medical images in immersive platforms. It remains unclear which VR input schemes and devices are optimal for these tasks. In this study, we perform a 12-person study to investigate user performance and experience while performing medical image segmentation with two control schemes, keyboard and mouse (KBM) and VR motion controllers (MC). Our results showed that motion controllers are faster in smaller segmentation tasks and offer a more pleasant user experience, however no significant difference in user accuracy was observed.

Index Terms: Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality, Interaction devices—Pointing devices, Interaction devices—Keyboards, Interaction devices—Haptic Devices

1 INTRODUCTION

Commercially available virtual reality (VR) and augmented reality (AR) head mounted displays (HMDs) have enabled immersive interactions in virtual duplicates of physical medical environments at a fraction of the cost[5]. Given the relative novelty of medical VR systems, uncertainties remain in the design of these systems, such as the optimal user input systems for specific clinical tasks such as image segmentation. Segmentation involves highlighting structures and areas of interest of specific anatomical elements to distinguish them from surrounding anatomy. While frequently highly automated, computer generated segmentation will frequently need to be manually modified to correct small errors or to highlight additional anatomy [1].

Segmentation is still predominantly performed using a 2D monitor with a keyboard and mouse(KBM) [4], however in VR systems the most readily available control systems are the system motion controllers(MC). [2]. In this study, we conducted a 12 participant user study that compares MC and KBM input modalities by examining participants' promptness, accuracy, and user experience in two segmentation tasks each comprised of 3 images.

2 METHODS

This study focuses on user performance in virtual-reality 2D segmentation tasks, which is the process of highlighting certain anatomical structures in a medical image. The a single 2D slice of the medical image onto 2D plane in the virtual world to better . The guide image was projected next to the markable image as seen in Figure (See figure 1).

Our study utilized two input schemes. The first, **Keyboard and Mouse(KBM)** utilized a keyboard and mouse for input in conjunction with a tethered VR headset to provide visualization. The user

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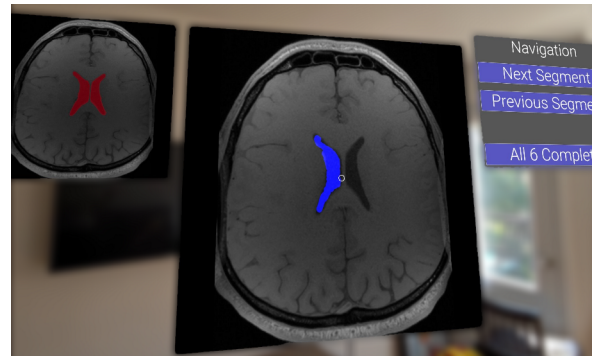


Figure 1: The user's viewpoint of the study, with the guide image on the left and the markable image in the center.

could move and zoom the image with the keyboard while the mouse was used exclusively to mark the image by controlling the X-Y positioning of a circular drawing cursor. Our second input scheme used **Motions Controllers (MC)** to provide input by having the right controller serve as a laser pointer to control the marking cursor, while the left controller was used to control image zooming and panning.

3 USER STUDY

The two tasks were designed to present different marking requirements to the participants. The first task (T1) required the segmentation the brain's ventricles, large smoothly edged structures in the center of the brain that would be conducive to a large cursor size and broad cursor strokes that segment large areas quickly. The second task(T2) required the segmentation of blood vessels in the brain, small branching structures in the brain that are best segmented using a small cursor size in short, curved motions that follow the center line of the vessel(see Figure 2).

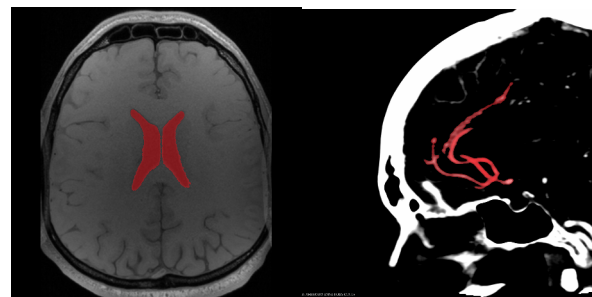


Figure 2: The T1 Ventricle Guide Image(Left) and the T2 Vessel Guide Image(Right). The participant was asked to mark the same areas on the active image as the areas in red on the guide image.

3.1 Hypotheses

Three hypotheses were put forward regarding the expected results of the study. Each hypothesis corresponds to a specific metric.

H1: MC will be faster for ventricle segmentation tasks (T1). Since this task requires fewer fine movements, the MC's freedom of motion in three axis will allow users to quickly mark large areas by allowing hand rotation and translation to move the cursor position. Meanwhile, KBM will be faster for the vessels task (T2) that requires finer curved movements due to the lack z-axis movement and rotation interfering with the smooth motion required, thus allowing the user to mark faster with continuous physical mouse movements.

H2: KBM will be more accurate for tasks T1 and T2. The increases in accuracy should come from the lack of rotational and Z-axis input with the mouse combined. Due to the lack of rotational and depth input, the participant can precisely move the cursor on the X-Y input axes that directly correspond to the X-Y coordinates of the segmentation surface.

H3: MC will be subjectively considered less physically and mentally demanding when compared to the KBM control scheme due to the low learning curve of a motion controller.

3.2 Procedures and Metrics

The test procedure consisted of the following three parts:

Pre-test Questionnaire and Tutorial: A pre-test questionnaire was administered to collect relevant demographic and background information about the participant. Demographic questions include the age, gender, and hand dominance.

Test: Participants were asked to complete the T1 and T2 tasks in both the KBM and MC control schemes. During the test, the accuracy of the user was measured by comparing how closely they matched the guide image. The time required to complete the task was also measured. Upon completion of both tasks, the user was prompted to take the post-test questionnaire.

Post-test Questionnaire: The post-test questionnaire was designed to capture the user's experience and preferences when using the motion controllers versus the mouse and keyboard using a 5-point Likert scale. It included the NASA TLX [3] questions regarding mental load, physical load, and temporal load for the T1 and T2 tasks for each control scheme, as well text entry feedback box.

4 RESULTS

The study was performed by 15 participants (7 male and 8 female), 3 of whom were removed due to significant deviations from the survey instructions or early termination of the study due to motion sickness. **Time** measurements ranged greatly (Min: 654sec, Max: 3785sec, Avg: 1656.56sec $\pm$ 864.13sec). In individual tasks, T1 showed no significant statistical difference between MC and KBM ($p = 0.101$), however MC proved to be faster in T2 ($p = 0.018$), thus partially refuting H1, where we initially hypothesized that KBM would outperform MC in fine motion tasks(H1). We believe that MC's freedom of motion on 3 axes allowed for more rapid targeting and segmenting of the desired areas. **Accuracy** measurements showed no significant differences between the tasks and their respective control schemes (MC T1:64.2% $\pm$ 14.56%, T2:61.62% $\pm$ 19.51%; KBM T1:69.96% $\pm$ 18.57, T2:64.17% $\pm$ 18.47), thus refuting H2. **Post-Test Questionnaire** refuted H3 by failing to show any statistically significance in how intuitive or demanding the control schemes were.(see Figure 3). User feedback indicated that both control schemes were demanding but that the MC control scheme

was more enjoyable. Future studies may examine if alternative motion controller schemes could reduce the demand-load while maintaining the enjoyment factor, and create an enjoyable, timely, and comfortable experience for the user.

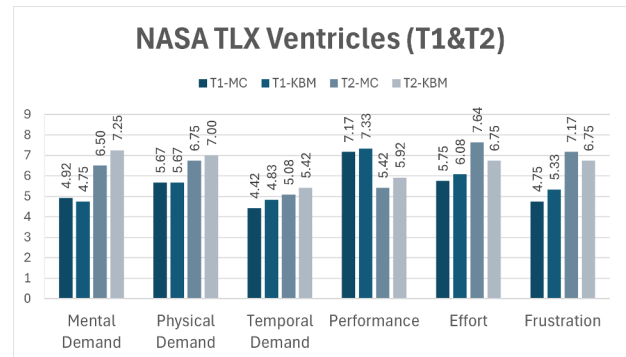


Figure 3: NASA TLX results for segmenting ventricles (T1) and vessels (T2).

5 CONCLUSION

Our study shows that while motion controllers are marginally faster vessel segmentation, they do not offer a significant advantage in accuracy over KBM, challenging the assumption that more intuitive controls would lead to improved performance. Our analysis of user feedback provides insights into how different control schemes affect user comfort and effectiveness, showing that selecting the proper input modality selection for medical image segmentation requires planning and understanding of the images and anatomical targets that will be segmented. These findings are valuable for the development of future VR-based medical tools, suggesting that task-specific input methods may be more beneficial than a one-size-fits-all approach.

ACKNOWLEDGMENTS

The questionnaire and methodology for this study was approved by the Concordia University Human Research and Ethics Committee under Protocol Number 30016074. This work was supported by Fonds Québécois de la Recherche sur la Nature et les Technologies, Grant Number VE0236.

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